

6 A $250 - 60 = 100 \ a$

$$a = 1.9 \text{ m s}^{-2}$$

$$T \cos 11^\circ = mg \text{ — (1)}$$

$$T \sin 11^\circ = ma \text{ — (2)}$$

$$\frac{(2)}{(1)}: \tan \theta = \frac{a}{g} = \frac{1.9}{9.81}$$

$$\Rightarrow \theta = 11^\circ$$